



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology Academic **Year 2025 - 2026 (Even Semester)**

Semester, Degree & Branch: II Semester, B.Tech & IT

Course Code & Title: IT25201& Foundations of Data Science using Python

Name of the Faculty member (s): Mrs. P.Ramya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Module II / Mathematical and Statistical Methods
- **Course Outcome:** CO2
- **Topic Learning Outcome:** -
- **Activity Chosen:** Think Share Pair
- **Justification:**

Choosing Think-Pair-Share for Mathematical and Statistical Methods is a highly effective pedagogical choice because it directly addresses the cognitive load and "math anxiety" often associated with these subjects. By incorporating a structured pause for individual reflection, students can engage in the productive struggle necessary to grasp complex formulas or data distributions without the pressure of immediate public performance. The subsequent peer discussion facilitates mathematical discourse, allowing students to test their logic, correct computational errors in a low-stakes environment, and discover multiple pathways to a solution. Ultimately, this method shifts the classroom from passive reception to active inquiry, ensuring that students do not just memorize procedures but also actually internalize the underlying logic and statistical reasoning.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - The time allotted for the topic is 10 Minutes.
 - Following the completion of Module 2, a 10-minute revision activity was conducted where students collaborated in groups to solve given two questions.(Boys and Girls Group)

Q1.You are provided with a NumPy array representing the daily production units of a factory over 7 days: `production = np.array([45, 52, 48, 102, 47, 50, 49])`

Write a Python/NumPy script to calculate the following and state the mathematical purpose of each. To calculate the following function `np.sum()`.

- a) The average daily output using `np.mean()`.
- b) The consistency of production using `np.std()`.
- c) The running total of production across the week using `np.cumsum()`.

Q2. Statistical Relationship [5 Marks]

In the dataset provided in Q1, the value 102 is an outlier.

- Calculate the `np.var()` and `np.std()` of the array.
- Justify: Mathematically, how can you derive the Standard Deviation if you only have the Variance?
- Analyze: If you replace 102 with the mean of the other six days, how will the `np.min()` and `np.max()` values change the perceived "range" of the data?

After Giving Questions students asked to do the followings:

- THINK (2 Mins):** Everyone works solo to solve the math.
Boys: Focus on writing the code for `np.mean()`, `np.std()`, and `np.cumsum()`.
Girls: Focus on the logic. How does replacing the outlier (102) change the range?
- PAIR (3 Mins):** Turn to your teammate. Compare your Python syntax and check your math.
Both team must agree on one thing: "Standard Deviation is just the square root of Variance."
- SHARE (5 Mins):** One representative from each team presents.
Boys shows the script on the board in Fig 1.
Fig 2. Shows during the activity
Girls explains why the "Range" of production drops from 57 to 7 once the outlier is gone.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO11	PSO1
CO1	3	3	3	3	3	2	3

(1 – Low 2 – Moderate 3 – High)

• PO/PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO11	PSO1
	3	3	3	3	3	2	3
Justification for Correlation	The students will apply knowledge of mathematics, science, and statistics fundamentals to solve complex problems	Students will be able to Identify, formulate, and analyze complex problems to reach substantiated conclusions using first principles of mathematics and statistics.	Students will be able to Design solutions for complex problems and processes that meet specified needs with appropriate consideration for societal and environmental factors	Students will be able to Use research-based methods, including the design of experiments and analysis/interpretation of data, to provide valid conclusions	Students will be able to Create, select, and apply appropriate techniques and modern tools (e.g., R, Python, MS Excel) for prediction and modeling.	The students will improve their knowledge in python continuously by implementing in emerging technologies	The students will have Acquire and apply core knowledge in major areas such as probability theory, statistical inference, and mathematical modeling to practical problems

- Images / Screenshot of the practice:

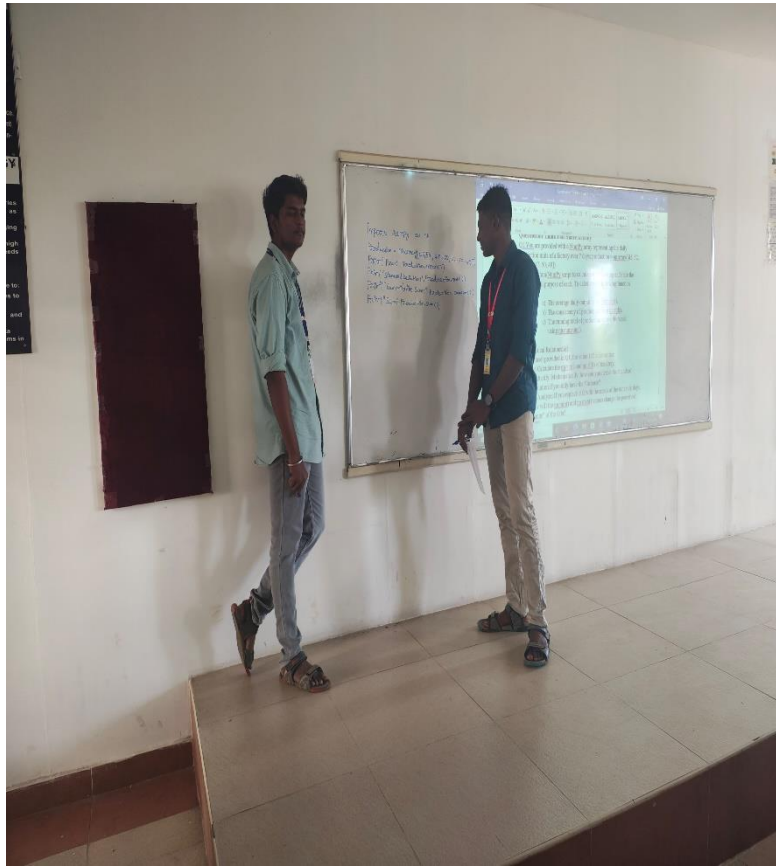


Figure 1. Presented by N.Vijaya Balaji and P.Tamilarasan



Figure 2. During the Activity

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Most students enjoyed the activity and participated actively in the discussion phases.
- Students easily explained simple terms like variables and syntax during the "Pair" and "Share" steps.
- Students found it harder to independently solve abstract logic and complex coding problems during the initial "Think" phase.

❖ *Benefit of the practice:*

- The "Think" phase helped students double-check their own knowledge of Python fundamentals.
- The "Pair" phase encouraged students to talk through problems together, improving their communication skills.
- The "Share" phase forced students to logically explain their answers to the class, boosting their analytical skills.

❖ *Challenges faced in implementation:*

- Discussions in the "Pair" phase often took longer than planned, making it hard to finish on time.
- Some students were very vocal, while others stayed quiet; making it difficult to ensure everyone participated.
- It was challenging to create problems that were "just right"—not too easy, but not so hard, that students were stuck.

References:

<https://www.ritrjpm.ac.in/images/B.Tech-IT/2025-2026/2.%20Think%20Pair%20Share.pdf>

https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/SS_IT3401_Validation.pdf

Signature of Faculty Member

HOD